

AIE1902: Large Language Models for Social Simulations

Assignment 2 — Reproducing a 3D Phase Diagram

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Background

In the paper Grauwis et al., “*Competition between collective and individual dynamics*” (PNAS, 2009), Fig. 3 shows a 3D phase diagram of the model’s behavior. The X-axis is the cooperativity parameter α , the Y-axis is the asymmetry parameter m , and the Z-axis is the average utility per agent U^* in the system’s final state.

This 3D surface reveals how the global outcome changes when agents move from purely individual decision-making ($\alpha = 0$) to purely collective decision-making ($\alpha = 1$), for different values of m . The flat “plateau” at high α corresponds to the mixed, optimal-utility phase; the lower area at small α corresponds to segregation.

In this assignment, you will reproduce a similar 3D diagram by running the simulation.

Learning Objectives

By completing this assignment, you will be able to:

- Use Google Colab to run a .ipynb notebook and generate a 3D visualization (surface plot) of simulation results.
- Interpret the resulting phase diagram and compare with published results.
- Document your workflow clearly and submit results in a reproducible format.

Your Tasks

1. 3D Surface Plot

- Plot α (X-axis), m (Y-axis), U^* (Z-axis) as a 3D surface.

2. Short Analysis (100–150 words) — include:

1. Group number; names & student IDs.
2. Describe the general pattern you see (answer the following points):
 - Where is U^* high? Where is it low?
 - What happens when α increases?
 - Is there a sharp transition between low and high U^* ?

3. Required Simulation Settings

Unless otherwise stated, use the following fixed settings:

- $Q=36$, $H=100$, $\rho_0=0.5$, $T=0$, $\text{max_steps} \geq 10000$

Deliverables

Submit a single ZIP named `AIE1902_A2_GroupX.zip` containing:

- 3D surface plot of U^* (in .jpg, .jpeg, or .png).
- Short analysis paragraph (100–150 words, in .pdf or .doc/.docx).
- (Optional but encouraged) A brief paragraph in your short analysis documenting any difficulties encountered and how you resolved them.

Format & Style Requirements

- Plots must be clear and easily recognisable.
- The short analysis should be written in clear English, concise and beginner-friendly.
- Submit in a ZIP archive with correct naming convention.

Assessment Rubric (100 pts)

Criterion	Description	Pts
Visualization	3D surface plot is clear and readable, with proper title and axis labels	50
Short Analysis	Analysis paragraph (100–150 words) is clear, concise, and insightful	30
Professionalism	Proper file naming, clean formatting, and well-organised documents	20

Academic Integrity & LLM Use

- Collaboration is within group only.
- You may use LLMs for help with coding and explanation, but you must verify all results and analysis.
- Do not copy full solutions directly from an LLM; the analysis must reflect your group's own work and understanding.
- Document any use of LLMs if any (e.g., prompts).

Submission

- **Where:** Blackboard